Problem 1 (Working with negligible functions). A non-negative function $\nu\colon \mathbb{N} \to \mathbb{R}$ is negligible if it decreases faster than the inverse of any polynomial; more precisely, for each polynomial P with coefficients in $\mathbb{R}$ (and non-negative leading coefficient), there exists some $N \in \mathbb{N}$ such that $\nu(n) < \frac{1}{P(n)}$ whenever $n > N$. Otherwise, we say that ν is non-negligible.

(i) Show that ν is negligible if and only if for every fixed sufficiently large integer c we have

$$\lim_{n\to\infty} \nu(n) \cdot n^c = 0.$$

(The displayed condition can be denoted by $\nu(n) = o(n^{-c})$.)

In the following, $\mathsf{negl}(n)$ denotes some arbitrary negligible function, and $\mathsf{poly}(n)$ denotes some arbitrary polynomial in n. State whether each of the following functions is negligible or non-negligible, and give a brief justification.

(i) $\nu(n) = 1/2^{100 \log n}$.

(ii) $\nu(n) = n^{-\log\log\log n}$.

(iii) $\nu(n) = \mathsf{poly}(n) \cdot \mathsf{negl}(n)$. (State whether ν is always negligible, or not necessarily.)

(iv) $\nu(n) = \mathsf{negl}(n)^{\frac{1}{\mathsf{poly}(n)}}$. (Same instructions as previous item.)

(v) Prove or disprove the following statement: $\nu(n)$ is non-negligible if and only if there exists a polynomial P with coefficients in $\mathbb{R}$ (and non-negative leading coefficient), there exists some $N \in \mathbb{N}$ such that $\nu(n) \geq \frac{1}{P(n)}$ whenever $n > N$.

(vi) Given finitely many tuples $(n_i, y_i) \in \mathbb{N} \times \mathbb{R}$ show that there are always a negligible function ν and a non-negligible function f such that $\nu(n_i) = f(n_i) = y_i$ for each i. Does the same conclusion still hold for infinitely many such tuples?

(vii) (I have no time to type in the description of this subproblem, sorry!)

(i). To prove the "if" part, suppose for every integer $c \geq N_0$ we have $\lim\limits_{n\to\infty} \nu(n)n^c = 0$. Now let $P(n) = a_k n^k + \ldots + a_1 n + a_0 \in \mathbb{R}[n]$ be an arbitrary polynomial with $k \in \mathbb{N}, a_k > 0$ (i.e. with non-negative leading coefficient) and define $c \triangleq \max(N_0, k)$. Note that for all

$$n > \left\lceil \max_{i=0,\ldots,k-1} \sqrt[k-i]{\frac{k|a_i|}{a_k}} \right\rceil \in \mathbb{N}$$

(denoted by N_1), we have the inequalities

$$\frac{a_k}{k} n^k > |a_i| n^i, \quad i = 0, \ldots, k-1, \tag{$*$}$$

which give us the following result:

$$P(n) = a_k n^k + \sum_{i=0,\ldots,k-1} a_i n^i \geq a_k n^k - \sum_{i=0,\ldots,k-1} |a_i| n^i = \sum_{i=0,\ldots,k-1} \left(\frac{a_k}{k} n^k - |a_i| n^i\right) \overset{(*)}{>} 0. \tag{$\clubsuit$}$$

We can then observe that for all $n > N_1$, $P(n)$ is in fact bounded by the monomial $2a_k n^c$ since

$$\begin{aligned}
P(n) &= a_k n^k + \sum_{i=0,\dots,k-1} a_i n^i \\
&\leq a_k n^k + \sum_{i=0,\dots,k-1} |a_i| n^i \\
&< a_k n^k + \sum_{i=0,\dots,k-1} \frac{a_k}{k} n^k && \text{using } (*) \\
&= 2a_k n^k \leq 2a_k n^c,
\end{aligned}$$

which implies

$$n^{-c} \cdot \frac{1}{2a_k} < \frac{1}{P(n)}. \tag{$\diamond$}$$

(Note also that in the above the properties that result in ($\diamond$): $\forall n > N_1 (0 < P(n) < 2a_k n^c)$ are actually derived by assuming $k > 0$. This is because if $k = 0$, it's trivial that $\forall n > N_1 (0 < P(n) = a_k < 2a_k n^c)$.)

On the other hand, since $\lim_{n\to\infty} \nu(n) n^c = 0$ and $\dfrac{1}{2a_k} > 0$, there exists some $N_2 \in \mathbb{N}$ such that

$$\nu(n) n^c = |\nu(n) n^c - 0| < \frac{1}{2a_k} \tag{$\circ$}$$

whenever $n > N_2$. (Note that $\nu(n)$ is non-negative.)

Hence, for all $n > \max(N_1, N_2) \in \mathbb{N}$, we have the inequality

$$\nu(n) \overset{(\circ)}{<} n^{-c} \cdot \frac{1}{2a_k} \overset{(\diamond)}{<} \frac{1}{P(n)}.$$

This shows that $\nu(n)$ is negligible by definition.

To prove the "only if" part, we assume $\nu(n)$ is negligible and let $c \in \mathbb{N}$. For every $\varepsilon > 0$, we know there exists some $N_\varepsilon \in \mathbb{N}$ such that $\nu(n) < \dfrac{1}{\varepsilon^{-1} n^c}$ whenever $n > N_\varepsilon$ as $\nu(n)$ is negligible and $\varepsilon^{-1} n^c \in \mathbb{R}[n]$ is a polynomial with non-negative leading coefficient, whence we have

$$|\nu(n) n^c - 0| = \nu(n) n^c < \varepsilon \quad \text{for all } n > N_\varepsilon$$

where the first equality holds since $\nu(n)$ is non-negative.

Thus, by definition, $\lim_{n\to\infty} \nu(n) n^c = 0$.

■

(ii). Non-negligible.

This is because for any integer $c \geq 100$,

$$\lim_{n\to\infty} \nu(n) n^c = \lim_{n\to\infty} \frac{n^c}{2^{100 \log n}} = \lim_{n\to\infty} \frac{n^c}{n^{100}} = 1 \text{ or } \infty,$$

implying $\nu(n) = \Omega(n^{-c})$. This implies $\nu(n)$ is non-negligible since we've shown in (i) that $\nu(n)$ is negligible iff $\nu(n) = o(n^{-c})$ for any sufficiently large integer c. ■

(iii). Negligible.
This is because for any $c \in \mathbb{N}$ and $\varepsilon_c > 0$,

$$\begin{aligned}|\nu(n)n^c - 0| = \nu(n)n^c = n^{c-\log\log\log n} & \\ < n^{c-(c+1)} & \qquad \text{since } n > 2^{2^{2^{c+1}}} \\ = n^{-1} & \\ < \varepsilon_c & \qquad \text{since } n > \varepsilon_c^{-1}\end{aligned}$$

whenever $n > \left\lceil \max\left(\varepsilon_c^{-1}, 2^{2^{2^{c+1}}}\right) \right\rceil \in \mathbb{N}$, which, by definition, leads to $\lim_{n\to\infty} \nu(n)n^c = 0$. This implies $\nu(n)$ is negligible since we've shown in (i) that $\nu(n)$ is negligible iff $\lim_{n\to\infty} \nu(n)n^c = 0$ for any sufficiently large integer c. ■

(iv). Negligible.
To prove the negligibility, we resort to the basic definition of it this time. Let $\nu(n) \triangleq R(n)\mu(n)$ where $R(n) \in \mathbb{R}[n]$ is a polynomial with non-negative leading coefficient and $\mu(n)$ be a negligible function. For each polynomial $P(n) \in \mathbb{R}[n]$ with non-negative leading coefficient, since $\mu(n)$ is negligible and $Q(n) \triangleq P(n)R(n) \in \mathbb{R}[n]$ is also a polynomial with non-negative leading coefficient, by definition there exists some $N \in \mathbb{N}$ such that $\mu(n) < \frac{1}{Q(n)}$ whenever $n > N$. It follows that for all $n > N$,

$$\nu(n) = R(n)\mu(n) < R(n) \cdot \frac{1}{Q(n)} = R(n) \cdot \frac{1}{P(n)R(n)} = \frac{1}{P(n)}.$$

This, by definition, implies $\nu(n)$ is also negligible. ■

(v). Not necessarily.
If $\nu(n) \triangleq (2^{-n^2})^{\frac{1}{n}} = 2^{-n}$, then $\nu(n)$ IS negligible since $\lim_{n\to\infty} \nu(n)n^c = \lim_{n\to\infty} \frac{n^c}{2^n} = 0$ for any integer $c \in \mathbb{N}$ and we've shown in (i) that $\nu(n)$ is negligible iff $\lim_{n\to\infty} \nu(n)n^c = 0$ for any sufficiently large integer c.
However, if $\nu(n) \triangleq (2^{-n^2})^{\frac{1}{n^2}} = \frac{1}{2}$, then $\nu(n)$ is NON-negligible since $\lim_{n\to\infty} \nu(n)n^c = \lim_{n\to\infty} \frac{n^c}{2} = \frac{1}{2}$ or $\infty \neq 0$ for all integer $c \in \mathbb{N}$ and we've shown in (i) that $\nu(n)$ is negligible iff $\lim_{n\to\infty} \nu(n)n^c = 0$ for any sufficiently large integer c. ■

(vi). A counterexample that violates the "only if" part: $\nu(n) \triangleq (-1)^n + 1$.
For the polynomial $P(n) \triangleq n$, it's clear that $\nu(n) = 2 \not< \frac{1}{n} = \frac{1}{P(n)}$ for all even and positive n. Since there are infinitely many such n, there is no $N \in \mathbb{N}$ such that $\nu(n) < \frac{1}{P(n)}$ whenever $n > N$. Thus $\nu(n)$ is indeed non-negligible.
On the other hand, for any polynomial $Q(n) \in \mathbb{R}[n]$ with non-negative leading coefficient, WLOG, suppose that $Q(x) = a_k n^k + \ldots + a_1 n + a_0$ and $k \in \mathbb{N}, a_k > 0$, and then it follows from the reasoning for (♣) that for all

$$n > \left\lceil \max_{i=0,\ldots,k-1} \sqrt[k-i]{\frac{k|a_i|}{a_k}} \right\rceil \in \mathbb{N}$$

(denoted by N'), we have

$$Q(n) \overset{(♣)}{>} 0.$$

Now observe that for all odd $n > N'$,

$$\nu(n) = (-1)^n + 1 = 0 < \frac{1}{Q(n)}.$$

Since there are infinitely many such n, there is no $N \in \mathbb{N}$ such that $\nu(n) \geq \frac{1}{Q(n)}$ whenever $n > N$. Hence the LHS holds while RHS does not. ■

(vii). Let ν' and f' be an arbitrary negligible function and an arbitrary non-negligible function, respectively (say, for instance, $\nu'(n) = 2^{-n}$ and $f'(n) = n^{-1}$).
For finitely many tuples, say $(n_i, y_i) \in \mathbb{N} \times \mathbb{R}, i = 0, 1, \ldots, k-1$, we simply define

$$\nu(n) \triangleq \begin{cases} y_i & n = n_i \text{ for some } i \\ \nu'(n) & \text{otherwise} \end{cases}, \quad f(n) \triangleq \begin{cases} y_i & n = n_i \text{ for some } i \\ f'(n) & \text{otherwise} \end{cases}.$$

Observe that $\nu(n) = f(n) = y_i$ for each i whereas ν and f are still negligible and non-negligible, respectively, just as ν' and f'. The reason is that the definition of negligibility only discusses the behavior of functions when n is sufficiently large. Precisely speaking, one can derive the negligibility of ν by replacing N by $\max(N, \max_{i=0,\ldots,k-1} n_i)$ in the definition of the negligibility of ν'; likewise, the non-negligibility of f' means that there is a polynomial $P(n)$ with coefficients in $\mathbb{R}$ and non-negative leading coefficient such that $f'(n) \geq \frac{1}{P(n)}$ at infinitely many integers n, which must also hold for f and thus f is also non-negligible.
However, this is not the case anymore when there are infinitely many tuples, say $(n_i, y_i) \in \mathbb{N} \times \mathbb{R}, i \in I$ where I is an infinite index set. Suppose $y_i = 1$ for every i and let ν be a negligible function. By definition, there is some $N \in \mathbb{N}$ such that $\nu(n) < \frac{1}{n}$ whenever $n > N$. This implies that $\nu(n_i) < \frac{1}{n_i} \leq 1 = y_i$ for any positive n_i $(i \in I)$. Since n_i's are distinct non-negative integers and I is an infinite set, there must be some $i \in I$ such that $n_i > 0$ and thus $\nu(n_i) \neq y_i$, which means the aforementioned conclusion does not hold anymore in this case. ■

Problem 2 (Large deviation bounds). Assume that $X_1, \dots, X_n$ are independent identically distributed (i.i.d.) random variables, each taking 1 with probability p and 0 with probability $1-p$. The Chernoff's bound says that for all $\varepsilon > 0$,

$$\Pr\left[\left|\frac{1}{n}\sum_i X_i - p\right| > \varepsilon\right] \leq 2e^{-n\varepsilon^2}.$$

If you are rusty on Chernoff's bound, read about it, e.g., here or search Google; there are lots of forms of the bound, the above being the most convenient for our applications.

(i) How large should n be if we want the average of the X_i to be within $\pm\varepsilon$ of p with probability at least $1-\delta$? (asymptotic expression for n is enough)

(ii) Imagine we used Chebyshev's bound instead of Chernoff's, and if you wish, assume for simplicity that $p = 1/2$. What bound on n would you get then? Do you see any advantage of Chebyshev's bound over Chernoff's?

(i). Observe that

$$\Pr\left[\left|\frac{1}{n}\sum_{i=0}^{n} X_i - p\right| \leq \varepsilon\right] = 1 - \Pr\left[\left|\frac{1}{n}\sum_{i=0}^{n} X_i - p\right| > \varepsilon\right] \geq 1 - 2e^{-n\varepsilon^2},$$

where the inequality is given by Chernoff's bound. Therefore, to arrive via Chernoff's bound at

$$\Pr\left[\left|\frac{1}{n}\sum_{i=0}^{n} X_i - p\right| \leq \varepsilon\right] \geq 1 - \delta,$$

the Chernoff's bound must be bounded below by $1-\delta$ first, i.e., $1 - 2e^{-n\varepsilon^2} \geq 1 - \delta$, from which we deduce the requirement for n:

$$1 - 2e^{-n\varepsilon^2} \geq 1 - \delta \iff 2e^{-n\varepsilon^2} \leq \delta \iff n \geq -\varepsilon^{-2}\ln(\frac{\delta}{2}) = \varepsilon^{-2}(\ln(\delta^{-1}) + \ln 2) = \Theta(\varepsilon^{-2}\ln(\delta^{-1})).$$

Hence n should be $O(\varepsilon^{-2}\ln(\delta^{-1}))$. ■

(ii). Since $X_1, \dots, X_n$ are independent and Bernoulli distributed with probability p, i.e., $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Ber}(p)$, we calculate the following:

$$E[X_i] = p, \qquad Var[X_i] = E[X_i^2] - E[X]^2 = p - p^2 = p(1-p),$$

$$E\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n} E[X_i] = p \qquad Var\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n^2}\sum_{i=1}^{n} Var[X_i] = \frac{p(1-p)}{n}.$$

Thus the Chebyshev's bound says that, for all $\varepsilon > 0$

$$\Pr\left[\left|\frac{1}{n}\sum_{i=0}^{n} X_i - E\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right]\right| > \varepsilon\right] \leq \frac{Var\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right]}{\varepsilon^2},$$

i.e.,

$$\Pr\left[\left|\frac{1}{n}\sum_{i=0}^{n} X_i - p\right| > \varepsilon\right] \leq \frac{p(1-p)}{n\varepsilon^2}.$$

Similar to the reasoning in the previous problem, observe that

$$\Pr\left[\left|\frac{1}{n}\sum_{i=0}^{n} X_i - p\right| \leq \varepsilon\right] = 1 - \Pr\left[\left|\frac{1}{n}\sum_{i=0}^{n} X_i - p\right| > \varepsilon\right] \geq 1 - \frac{p(1-p)}{n\varepsilon^2},$$

where the inequality is given by Chebyshev's bound. Therefore, to arrive via Chebyshev's bound at

$$\Pr\left[\left|\frac{1}{n}\sum_{i=0}^{n} X_i - p\right| \leq \varepsilon\right] \geq 1 - \delta,$$

the Chebyshev's bound must be bounded below by $1 - \delta$ first, i.e., $1 - \frac{p(1-p)}{n\varepsilon^2} \geq 1 - \delta$, from which we deduce the requirement for n:

$$1 - \frac{p(1-p)}{n\varepsilon^2} \geq 1 - \delta \iff \frac{p(1-p)}{n\varepsilon^2} \leq \delta \iff n \geq p(1-p)\delta^{-1}\varepsilon^{-2} \overset{\text{if } p=\frac{1}{2}}{=} \frac{1}{4}(\delta^{-1}\varepsilon^{-2}).$$

Hence n should be no less than $p(1-p)\delta^{-1}\varepsilon^{-2}$, or $n = O(\delta^{-1}\varepsilon^{-2})$ if p is constant.

We see some advantages of Chernoff's bound over Chebyshev's here:

1. Chernoff's bound is tighter than Chebyshev's, as the former decreases exponentially, being much faster than the latter, which decreases polynomially, both as functions of n. Meanwhile, the lower bound for n deduced from Chernoff's bound grows much slower and is thus tighter than that deduced from Chebyshev's: the former gives $O(\varepsilon^{-2}\ln(\delta^{-1}))$ while the latter gives $O(\delta^{-1}\varepsilon^{-2})$ (supposing p is a constant).

2. Chernoff's lower bound for n is independent of p, which is great compared to Chebyshev's!

As to the advantage of Chebyshev's bound over Chernoff's, I think one is that Chebyshev's bound, though being looser, requires less assumptions compared to Chernoff's (e.g. Chernoff's requires that $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Ber}(p)$) and thus more general, that is, it can be applied in more settings as long as the mean and variance of the random variable. ■

Problem 3 (Union bounds). Let $A, B_1, \dots, B_k$ be some arbitrary probabilistic events.

(i) Show that

$$\Pr[B_1 \cup B_2 \cup \dots \cup B_k] \le \sum_{i=1}^{k} \Pr[B_k].$$

(ii) Show that

$$\Pr[A \cap (B_1 \cup B_2 \cup \dots \cup B_k)] \le \sum_{i=1}^{k} \Pr[A \mid B_k].$$

(iii) Let F be the set of functions mapping $\{1, 2, \dots, n\}$ to $\{1, 2, \dots, m\}$, where $m \ge n^2$. Show that

$$\Pr_{f \leftarrow F}[f \text{ is injective}] \ge 1 - \frac{n^2}{2m}.$$

Specifically, we use $f \leftarrow F$ to denote that f is chosen uniformly at random from F and a function is injective if $f(x) \ne f(y)$ for any distinct x, y.

(i). Define $B_1^* \triangleq B_1$ and $B_i^* \triangleq B_i \setminus \bigcup_{j=1}^{i-1} B_j$ for each $i = 2, 3 \dots, k$. Notice that $B_i^* \cap B_j^* = \varnothing \; \forall i \ne j$ and that $B_i^* \subseteq B_i \; \forall i$.

Claim 3.1. $\bigcup_{i=1}^{k} B_i = \bigcup_{i=1}^{k} B_i^*$

Proof. Let $\omega \in \bigcup_{i=1}^{k} B_i$. Then there must be some $1 \le n \le k$ such that $\omega \in B_n$. By the well-ordering of $\mathbb{Z}$, choose the smallest n_0 such that $\omega \in B_{n_0}$. Equivalently,

$$\omega \notin B_i \text{ for all } i = 1, \dots, n_0 - 1 \quad \text{and} \quad \omega \in B_{n_0}.$$

Hence,

$$\omega \in B_{n_0} \setminus \bigcup_{i=1}^{n_0 - 1} B_i = B_{n_0}^* \subseteq \bigcup_{i=1}^{k} B_i^*,$$

which means $\bigcup_{i=1}^{k} B_i \subseteq \bigcup_{i=1}^{k} B_i^*$.
Conversely, let $\omega \in \bigcup_{i=1}^{k} B_i^*$. Then there must be some $1 \le n \le k$ such that $\omega \in B_n^*$. Thus

$$\omega \in B_n^* = B_n \setminus \bigcup_{i=1}^{n-1} B_i \subseteq B_n \subseteq \bigcup_{i=1}^{k} B_i,$$

which means $\bigcup_{i=1}^{k} B_i \supseteq \bigcup_{i=1}^{k} B_i^*$. ∎

Hence,

$$\begin{aligned}
\Pr[B_1 \cup B_2 \cup \dots \cup B_k] &= \Pr\left[\bigcup_{i=1}^{k} B_i\right] \\
&= \Pr\left[\bigcup_{i=1}^{k} B_i^*\right] && \text{by the claim} \\
&= \sum_{i=1}^{k} \Pr[B_i^*] && \text{since } B_i^* \cap B_j^* = \varnothing \; \forall i \ne j \\
&\le \sum_{i=1}^{k} \Pr[B_i] && \text{since } B_i^* \subseteq B_i \implies \Pr[B_i^*] \le \Pr[B_i] \; \forall i.
\end{aligned}$$

∎

(ii).

$$\begin{aligned}
\Pr[A \cap (B_1 \cup B_2 \cup \cdots \cup B_k)] &= \Pr[(A \cap B_1) \cup (A \cap B_2) \cup \cdots \cup (A \cap B_k)] \\
&\le \sum_{i=1}^{k} \Pr[A \cap B_i] && \text{using the result of (i)} \\
&= \sum_{i=1}^{k} \Pr[B_i] \Pr[A \mid B_i] \\
&\le \sum_{i=1}^{k} \Pr[A \mid B_i] && \text{since } \Pr[B_i] \le 1 \; \forall i
\end{aligned}$$

■

(iii). Let F be the sample space and denote by E_{xy} the event in which $f(x) = f(y)$ for each $x, y \in \{1, 2, \ldots, n\}$, that is,

$$E_{xy} \triangleq \{f \in F \mid f(x) = f(y)\}.$$

Note that since f is uniformly chosen from F, for any distinct $x, y \in \{1, 2, \ldots, n\}$,

$$\Pr_{f \leftarrow F}[E_{xy}] = \frac{|E_{xy}|}{|F|} = \frac{m \cdot m^{n-2}}{m^n} = \frac{1}{m}. \tag{♠}$$

This leads us to an upper bound for $\Pr_{f \leftarrow F}[f \text{ is not injective}]$:

$$\begin{aligned}
\Pr_{f \leftarrow F}[f \text{ is not injective}] &= \Pr_{f \leftarrow F}\left[\bigcup_{1 \le x < y \le n} E_{xy}\right] && \text{by the definition of injectivity} \\
&\le \sum_{1 \le x < y \le n} \Pr_{f \leftarrow F}[E_{xy}] && \text{using the result of (i)} \\
&= \sum_{1 \le x < y \le n} \frac{1}{m} && \text{by (♠)} \\
&= \frac{1}{m} \cdot \binom{n}{2} = \frac{n(n-1)}{2m} \le \frac{n^2}{2m}. && (\bigstar)
\end{aligned}$$

Hence,

$$\Pr_{f \leftarrow F}[f \text{ is injective}] = 1 - \Pr_{f \leftarrow F}[f \text{ is not injective}] \overset{(\bigstar)}{\ge} 1 - \frac{n^2}{2m}.$$

■

Problem 4 (One-Way Functions). We first review the concept of one-way functions which we've introduced in the lecture. A function $f\colon \{0,1\}^* \to \{0,1\}^*$ is one-way if it can be efficiently computed, but no efficient algorithm can invert it. More precisely, for any probabilistic polynomial time (PPT) algorithm $\mathbf{Adv}$,

$$\Pr_{x \xleftarrow{\$} \{0,1\}^n}[f(\mathbf{Adv}(1^n, f(x)) = f(x)] = \mathsf{negl}(n)$$

In this problem, we focus on one-way function f that is length-preserving, i.e., for every $x \in \{0,1\}^*$ it holds that $|f(x)| = |x|$. Note that $|$ denotes concatenation, e.g., $1011|0000 = 10110000$, $\oplus$ denotes bit-by-bit XOR, $\wedge$ denotes bit-by-bit AND, and $x_{[a:b]}$ denotes the substring of x from the ath bit to the bth bits.

For each of the following functions g, assuming f is a one-way function, prove that g is a one-way function, or provide a counter example to demonstrate that it is not always a one-way function. More specifically:

- In the case that g is one-way, you should show a reduction from inverting f to inverting g. In other word, you should show that if g is efficiently invertible, then f is also efficiently invertible.
- In the case that g is not always one-way, you should do the following. Assuming only that length-preserving one-way functions exist, you should construct a length-preserving one-way function f such that when instantiated with f, g is not a one-way function. That is, you should show a polynomial-time inverting algorithm for g.

For example, to prove that the construction of $g(x) = f(x_{[1:n/2]}|0^{n/2})$ is not always one-way, the argument goes as follows. Assume that h is an arbitrary length-preserving one-way function. Construct the function

$$f(x) = \begin{cases} 0^n & x_{[n/2+1,n]} = 0^{n/2} \\ h(x) & \text{otherwise.} \end{cases}$$

f is one-way assuming that h is one-way (you have to show this by a reduction). Furthermore, g constructed using this f is not one-way (you have to show this by exhibiting a polynomial-time attack). In general, your solutions should contain complete proofs that the counter examples are valid. In the following, assume $|x| = n$,

(i) $g(x) = f(x_{[1:\frac{n}{2}]})|x_{[\frac{n}{2}+1:n]}$

(ii) $g(x) = f(x)|f(f(x))$

(iii) $g(x) = f(f(x))$ where f is a one-way permutation, i.e., f is one-way and bijective.

(iv) $g(x) = f(x) \oplus x$

(v) $g(x|x') = f(x \wedge x')$ for all $x, x' \in \{0,1\}^n$. Here $g : \{0,1\}^{2n} \to \{0,1\}^n$.

For the following subproblems we will prove a trivial property first:

Lemma 4.1. Let $\nu, \mu : \mathbb{N} \to \mathbb{R}$ be non-negative functions such that there exists $N \in \mathbb{N}$ such that $\nu(n) \geq \mu(n)$ for all $n > N$. If μ is non-negligible, then ν is also non-negligible.

Proof of Lemma 4.1. We prove this by contradiction.
Suppose to the contrary that ν is negligible, which implies from the conclusion we have in problem

1(i) that for every fixed sufficiently large integer c we have

$$\lim_{n\to\infty} \nu(n)n^c = 0;$$

equivalently, for any $\varepsilon_c > 0$ there is $N_{\varepsilon_c} \in \mathbb{N}$ such that $\nu(n)n^c = |\nu(n)n^c - 0| < \varepsilon_c$ whenever $n > N_{\varepsilon_c}$. Therefore, for every fixed sufficiently large integer c and any $\varepsilon_c > 0$, we have

$$|\mu(n)n^c - 0| = \mu(n)n^c \le \nu(n)n^c < \varepsilon_c$$

whenever $n > \max(N_{\varepsilon_c}, N) \in \mathbb{N}$. Equivalently,

$$\lim_{n\to\infty} \mu(n)n^c = 0.$$

for every fixed sufficiently large integer c. Hence μ must be negligible as well, which is contradictory to the antecedent. ■

Lemma 4.2. Let $\nu, \mu : \mathbb{N} \to \mathbb{R}$ be non-negative functions such that there exists $N \in \mathbb{N}$ such that $\nu(n) \ge \mu(n)$ for all $n > N$. If ν is negligible then so is μ.

Proof of Lemma 4.2. This is simply the contrapositive of lemma 4.1. ■

(i). We prove g is one-way via reduction.
Suppose to the contrary that g is not one-way, i.e., there exists a PPT algorithm $\mathbf{Adv}$ such that

$$\varepsilon(n) \triangleq \Pr_{x \xleftarrow{\$} \{0,1\}^n} [g(\mathbf{Adv}(1^n, g(x))) = g(x)]$$

is non-negligible. That is to say, $\mathbf{Adv}$ efficiently inverts g with non-negligible probability. (Note that we've ignored the scenario where g is neither efficiently invertible nor efficiently computable. This is however impossible here since f can be computed in PPT and so can g.) Our goal is to construct another PPT algorithm $\mathbf{Adv}^*$ that inverts f with non-negligible probability. Since f is one-way by assumption, this is a contradiction.

We construct $\mathbf{Adv}^*$ as follows: $\mathbf{Adv}^*$ takes as input 1^m and $y := f(x)$, where $x \leftarrow \{0,1\}^m$. Then, it uniformly samples $s \leftarrow \{0,1\}^m$ and runs $\mathbf{Adv}$ on input $(1^{2m}, y|s)$, receiving as output x'. Finally, it outputs $x'_{[1:m]}$. Namely,

$$\mathbf{Adv}^*(1^m, y) \triangleq \mathbf{Adv}(1^{2m}, y|s)_{[1:m]} \quad \text{where} \quad s \xleftarrow{\$} \{0,1\}^m.$$

The following diagram illustrates the algorithm we've just constructed.

Challenger		$\mathbf{Adv}^*$		$\mathbf{Adv}$
sample $x \xleftarrow{\$} \{0,1\}^m$				
	$\xrightarrow{(1^m, y:=f(x))}$			
		sample $s \xleftarrow{\$} \{0,1\}^m$		
			$\xrightarrow{(1^{2m}, y\|s)}$	
				$[\cdots]$
			$\xleftarrow{x'}$	
	$\xleftarrow{x'_{[1:m]}}$			

It suffices to verify that $\mathbf{Adv}^*$ is a PPT algorithm that inverts f with non-negligible probability. Firstly, it's clear that $\mathbf{Adv}^*$ runs in PPT as it simply samples m bits, concatenates bit strings once, runs $\mathbf{Adv}$ on a $2m$-bit input, which also runs in PPT, and splits a bit string. For the second part, we claim that

$$\Pr_{x \xleftarrow{\$} \{0,1\}^m} [f(\mathbf{Adv}^*(1^m, f(x))) = f(x)]$$

is non-negligible. To see this, observe that for any $m \in \mathbb{N}$

$$\begin{aligned}
\Pr_{x \xleftarrow{\$} \{0,1\}^m} [f(\mathbf{Adv}^*(1^m, f(x))) = f(x)] &= \Pr_{x,s \xleftarrow{\$} \{0,1\}^m} \left[f(\mathbf{Adv}(1^{2m}, f(x)|s)_{[1:m]}) = f(x)\right] \\
&\geq \Pr_{x,s \xleftarrow{\$} \{0,1\}^m} \left[g(\mathbf{Adv}(1^{2m}, f(x)|s)) = f(x)|s\right] \\
&= \Pr_{x^* \xleftarrow{\$} \{0,1\}^{2m}} \left[g(\mathbf{Adv}(1^{2m}, f(x^*_{[1,m]})|x^*_{[m+1,2m]})) = f(x^*_{[1,m]})|x^*_{[m+1,2m]}\right] \\
&\qquad \text{(since } x|s \text{ is distributed identically to a uniformly random } 2m\text{-bit string)} \\
&= \Pr_{x^* \xleftarrow{\$} \{0,1\}^{2m}} \left[g(\mathbf{Adv}(1^{2m}, g(x^*))) = g(x^*)\right] \\
&\qquad \text{(by def. of } g\text{)} \\
&= \varepsilon(2m),
\end{aligned}$$

where the inequality comes from the fact that

$$\begin{aligned}
g(\mathbf{Adv}(1^{2m}, f(x)|s)) = f(x)|s \iff & f(\mathbf{Adv}(1^{2m}, f(x)|s)_{[1:m]})|\mathbf{Adv}(1^{2m}, f(x)|s)_{[m+1,2m]} = f(x)|s \\
\implies & f(\mathbf{Adv}(1^{2m}, f(x)|s)_{[1:m]}) = f(x).
\end{aligned}$$

As $\varepsilon(2m)$ is non-negligible, which follows from the fact that $\varepsilon(m)$ is non-negligible, by lemma 4.1,

$$\Pr_{x \xleftarrow{\$} \{0,1\}^m} [f(\mathbf{Adv}^*(1^m, f(x))) = f(x)]$$

is also non-negligible. This proves that $\mathbf{Adv}^*$ runs in PPT and inverts f with non-negligible probability, contradicting the assumption that f is one-way. ■

(ii). We prove g is one-way via reduction.
Suppose to the contrary that g is not one-way, i.e., there exists a PPT algorithm $\mathbf{Adv}$ such that

$$\varepsilon(n) \triangleq \Pr_{x \xleftarrow{\$} \{0,1\}^n} [g(\mathbf{Adv}(1^n, g(x))) = g(x)]$$

is non-negligible, i.e., that $\mathbf{Adv}$ efficiently inverts g with non-negligible probability. (Note that we've ignored the scenario where g is neither efficiently invertible nor efficiently computable. This is however impossible here since f can be computed in PPT and so can g.) Our goal is to construct another PPT algorithm $\mathbf{Adv}^*$ that inverts f with non-negligible probability. Since f is one-way by assumption, this is a contradiction.

We construct $\mathbf{Adv}^*$ as follows: $\mathbf{Adv}^*$ takes as input 1^m and $y := f(x)$, where $x \leftarrow \{0,1\}^m$. Then, it runs $\mathbf{Adv}$ on input $(1^m, y|f(y))$, receiving as output x'. Finally, it simply outputs x'. Namely,

$$\mathbf{Adv}^*(1^m, y) \triangleq \mathbf{Adv}(1^m, y|f(y)).$$

The following diagram illustrates the algorithm we've just constructed.

$$
\begin{array}{ccccc}
\text{Challenger} & & \mathbf{Adv}^* & & \mathbf{Adv} \\
\text{sample } x \xleftarrow{\$} \{0,1\}^m & & & & \\
& \xrightarrow{(1^m, y := f(x))} & & & \\
& & z := y | f(y) & & \\
& & & \xrightarrow{(1^m, z)} & \\
& & & & [\cdots] \\
& & & \xleftarrow{x'} & \\
& \xleftarrow{x'} & & &
\end{array}
$$

It suffices to verify that $\mathbf{Adv}^*$ is a PPT algorithm that inverts f with non-negligible probability. Firstly, it's clear that $\mathbf{Adv}^*$ runs in PPT as it simply computes f, which is computable in PPT as f is one-way by assumption, concatenates bit strings once, and then runs $\mathbf{Adv}$ on a $2m$-bit input, which also runs in PPT. For the second part, we claim that

$$\Pr_{x \xleftarrow{\$} \{0,1\}^m} [f(\mathbf{Adv}^*(1^m, f(x))) = f(x)]$$

is non-negligible. To see this, observe that for any $m \in \mathbb{N}$

$$
\begin{aligned}
\Pr_{x \xleftarrow{\$} \{0,1\}^m} [f(\mathbf{Adv}^*(1^m, f(x))) = f(x)] &= \Pr_{x \xleftarrow{\$} \{0,1\}^m} [f(\mathbf{Adv}(1^m, f(x)|f(f(x)))) = f(x)] \\
&\overset{\text{(a)}}{=} \Pr_{x \xleftarrow{\$} \{0,1\}^m} [g(\mathbf{Adv}(1^m, f(x)|f(f(x)))) = f(x)|f(f(x))] \\
&= \Pr_{x \xleftarrow{\$} \{0,1\}^m} [g(\mathbf{Adv}(1^m, g(x)) = g(x)] \\
&= \varepsilon(m),
\end{aligned}
$$

where equality (a) comes from the fact that

$$
\begin{aligned}
& g(\mathbf{Adv}(1^m, f(x)|f(f(x)))) = f(x)|f(f(x)) \\
\iff\quad & f(\mathbf{Adv}(1^m, f(x)|f(f(x))))|f(f(\mathbf{Adv}(1^m, f(x)|f(f(x))))) = f(x)|f(f(x)) \\
\iff\quad & f(\mathbf{Adv}(1^m, f(x)|f(f(x)))) = f(x).
\end{aligned}
$$

As $\varepsilon(m)$ is non-negligible by assumption,

$$\Pr_{x \xleftarrow{\$} \{0,1\}^m} [f(\mathbf{Adv}^*(1^m, f(x))) = f(x)]$$

is also non-negligible. This proves that $\mathbf{Adv}^*$ runs in PPT and inverts f with non-negligible probability, contradicting the assumption that f is one-way. ∎

(iii). We prove g is one-way via reduction.
Suppose to the contrary that g is not one-way, i.e., there exists a PPT algorithm $\mathbf{Adv}$ such that

$$\varepsilon(n) \triangleq \Pr_{x \xleftarrow{\$} \{0,1\}^n} [g(\mathbf{Adv}(1^n, g(x))) = g(x)]$$

is non-negligible, i.e., that $\mathbf{Adv}$ efficiently inverts g with non-negligible probability. (Note that we've ignored the scenario where g is neither efficiently invertible nor efficiently computable. This is however impossible here since f can be computed in PPT and so can g.) Our goal is to construct another PPT algorithm $\mathbf{Adv}^*$ that inverts f with non-negligible probability. Since f is one-way by assumption, this is a contradiction.

We construct $\mathbf{Adv}^*$ as follows: $\mathbf{Adv}^*$ takes as input 1^m and $y := f(x)$, where $x \leftarrow \{0,1\}^m$. Then, it computes $z := f(y)$ and runs $\mathbf{Adv}$ on input $(1^m, z)$, receiving as output x'. Finally, it simply outputs x'. Namely,

$$\mathbf{Adv}^*(1^m, y) \triangleq \mathbf{Adv}(1^m, f(y)).$$

The following diagram illustrates the algorithm we've just constructed.

$$\begin{array}{ccccc}
\text{Challenger} & & \mathbf{Adv}^* & & \mathbf{Adv} \\
\text{sample } x \xleftarrow{\$} \{0,1\}^m & & & & \\
& \xrightarrow{(1^m, y := f(x))} & & & \\
& & z := f(y) & & \\
& & & \xrightarrow{(1^m, z)} & \\
& & & & [\cdots] \\
& & & \xleftarrow{x'} & \\
& \xleftarrow{x'} & & &
\end{array}$$

It suffices to verify that $\mathbf{Adv}^*$ is a PPT algorithm that inverts f with non-negligible probability. Firstly, it's clear that $\mathbf{Adv}^*$ runs in PPT as it simply computes f, which is computable in PPT as f is one-way by assumption, and then runs $\mathbf{Adv}$ on an m-bit input, which also runs in PPT. For the second part, we claim that

$$\Pr_{x \xleftarrow{\$} \{0,1\}^m} [f(\mathbf{Adv}^*(1^m, f(x))) = f(x)]$$

is non-negligible. To see this, observe that for any $m \in \mathbb{N}$

$$\begin{aligned}
\Pr_{x \xleftarrow{\$} \{0,1\}^m} [f(\mathbf{Adv}^*(1^m, f(x))) = f(x)] &= \Pr_{x \xleftarrow{\$} \{0,1\}^m} [f(\mathbf{Adv}(1^m, f(f(x)))) = f(x)] \\
&\overset{\text{(a)}}{=} \Pr_{x \xleftarrow{\$} \{0,1\}^m} [g(\mathbf{Adv}(1^m, f(f(x)))) = f(f(x))] \\
&= \Pr_{x \xleftarrow{\$} \{0,1\}^m} [g(\mathbf{Adv}(1^m, g(x))) = g(x)] \\
&= \varepsilon(m),
\end{aligned}$$

where equality (a) comes from the fact that f is injective (as it is a permutation by assumption) and therefore

$$\begin{aligned}
g(\mathbf{Adv}(1^m, f(f(x)))) = f(f(x)) &\overset{\text{definition of } g}{\iff} f(f(\mathbf{Adv}(1^m, f(f(x))))) = f(f(x)) \\
&\overset{\text{injectivity of } f}{\iff} f(\mathbf{Adv}(1^m, f(f(x)))) = f(x).
\end{aligned}$$

As $\varepsilon(m)$ is non-negligible by assumption,

$$\Pr_{x \xleftarrow{\$} \{0,1\}^m} [f(\mathbf{Adv}^*(1^m, f(x))) = f(x)]$$

is also non-negligible. This proves that $\mathbf{Adv}^*$ runs in PPT and inverts f with non-negligible probability, contradicting the assumption that f is one-way. ■

(iv). We show g is not always one-way by providing a counterexample, i.e., a length-preserving one-way function f such that the corresponding g is not one-way.

Let h be an arbitrary length-preserving one-way function (which exists by the assumption that a length-preserving one-way function f exists). We define our $f : \{0,1\}^* \to \{0,1\}^*$ as

$$f(x) \triangleq h(x_{[\frac{n}{2}+1,n]})|0^{\frac{n}{2}} \quad \text{where } n \text{ denotes } |x|,$$

which we claim to be length-preserving and one-way.

Claim 4.1. f is a length-preserving one-way function.

Proof. Firstly, it's immediate that f is length-preserving as h is as well. Notice that f can be computed in PPT since h is one-way and thus can be computed in PPT by definition. Then, we prove f is one-way via reduction.

Suppose to the contrary that f is not one-way. As f can be computed in PPT, it must be that there exists a PPT algorithm $\mathbf{Adv}$ such that

$$\varepsilon(n) \triangleq \Pr_{x \xleftarrow{\$} \{0,1\}^n} [f(\mathbf{Adv}(1^n, f(x))) = f(x)]$$

is non-negligible. Then, we construct another PPT algorithm $\mathbf{Adv}^*$ that inverts h with non-negligible probability. Since h is one-way by assumption, this is a contradiction.

We construct $\mathbf{Adv}^*$ as follows: $\mathbf{Adv}^*$ takes as input 1^m and $y \coloneqq h(x)$, where $x \leftarrow \{0,1\}^m$. Then, it runs $\mathbf{Adv}$ on input $(1^{2m}, y|0^m)$, receiving as output x', and finally outputs $x'_{[m+1,2m]}$. Namely,

$$\mathbf{Adv}^*(1^m, y) \triangleq \mathbf{Adv}(1^{2m}, y|0^m)_{[m+1,2m]}.$$

It suffices to verify that $\mathbf{Adv}^*$ is a PPT algorithm that inverts h with non-negligible probability. Firstly, it's clear that $\mathbf{Adv}^*$ runs in PPT as it simply concatenates bit strings, runs $\mathbf{Adv}$ on a $2m$-bit input, which also runs in PPT, and then splits a bit string. For the second part, we claim

$$\Pr_{x \xleftarrow{\$} \{0,1\}^m} [h(\mathbf{Adv}^*(1^m, h(x))) = h(x)]$$

is non-negligible. To see this, observe that for any $m \in \mathbb{N}$

$$\begin{aligned}
\Pr_{x \xleftarrow{\$} \{0,1\}^m} [h(\mathbf{Adv}^*(1^m, h(x))) = h(x)] &= \Pr_{x \xleftarrow{\$} \{0,1\}^m} \left[h(\mathbf{Adv}(1^{2m}, h(x)|0^m)_{[m+1,2m]}) = h(x)\right] \\
&\overset{\text{(a)}}{=} \Pr_{x \xleftarrow{\$} \{0,1\}^m} \left[f(\mathbf{Adv}(1^{2m}, h(x)|0^m)) = h(x)|0^m\right] \\
&= \Pr_{x^* \xleftarrow{\$} \{0,1\}^{2m}} \left[f(\mathbf{Adv}(1^{2m}, h(x^*_{[m+1,2m]})|0^m)) = h(x^*_{[m+1,2m]})|0^m\right] \\
&\qquad \text{(since } x \text{ is distributed identically to the last } m \text{ bits of } x^* \xleftarrow{\$} \{0,1\}^{2m}) \\
&= \Pr_{x^* \xleftarrow{\$} \{0,1\}^{2m}} \left[f(\mathbf{Adv}(1^{2m}, f(x^*))) = f(x^*)\right] = \varepsilon(2m),
\end{aligned}$$

where equality (a) comes from the fact that

$$\begin{aligned}
f(\mathbf{Adv}(1^{2m}, h(x)|0^m)) = h(x)|0^m &\iff h(\mathbf{Adv}(1^{2m}, h(x)|0^m)_{[m+1,2m]})|0^m = h(x)|0^m \\
&\iff h(\mathbf{Adv}(1^{2m}, h(x)|0^m)_{[m+1,2m]}) = h(x).
\end{aligned}$$

As $\varepsilon(m)$ is non-negligible by assumption and so is $\varepsilon(2m)$,

$$\Pr_{x \xleftarrow{\$} \{0,1\}^m} [h(\mathbf{Adv}^*(1^m, h(x))) = h(x)]$$

is also non-negligible. This proves that $\mathbf{Adv}^*$ runs in PPT and inverts h with non-negligible probability, contradicting the assumption that h is one-way. ■

Next, to show g is not one-way given the f we've just defined, we provide a PPT algorithm $\mathbf{Adv}^\bigstar$ that inverts g with non-negligible probability, i.e., that

$$\nu(n) \triangleq \Pr_{x \xleftarrow{\$} \{0,1\}^n} \left[g(\mathbf{Adv}^\bigstar(1^n, g(x))) = g(x) \right]$$

is non-negligible. We construct $\mathbf{Adv}^\bigstar$ as follows: $\mathbf{Adv}^*$ takes as input 1^n and $y \coloneqq g(x)$, where $x \leftarrow \{0,1\}^n$. Then, it outputs $h(y_{[\frac{n}{2}+1,n]}) \oplus y_{[1,\frac{n}{2}]} | y_{[\frac{n}{2}+1,n]}$. Namely,

$$\mathbf{Adv}^\bigstar(1^n, y) \triangleq h(y_{[\frac{n}{2}+1,n]}) \oplus y_{[1,\frac{n}{2}]} | y_{[\frac{n}{2}+1,n]}.$$

It suffices to verify that $\mathbf{Adv}^\bigstar$ is a PPT algorithm that inverts g with non-negligible probability. Firstly, it's clear that $\mathbf{Adv}^\bigstar$ runs in PPT as it simply computes h, which is computable in PPT as it is one-way by assumption, taking XOR, and concatenates bit strings together. For the second part, observe that for any $n \in \mathbb{N}$

$$\begin{aligned}
\nu(n) &\triangleq \Pr_{x \xleftarrow{\$} \{0,1\}^n} \left[g(\mathbf{Adv}^\bigstar(1^n, g(x)) = g(x) \right] \\
&= \Pr_{x \xleftarrow{\$} \{0,1\}^n} \left[g(h(g(x)_{[\frac{n}{2}+1,n]}) \oplus g(x)_{[1,\frac{n}{2}]} | g(x)_{[\frac{n}{2}+1,n]}) = g(x) \right] \\
&= \Pr_{x \xleftarrow{\$} \{0,1\}^n} \left[g(h(y_{[\frac{n}{2}+1,n]}) \oplus y_{[1,\frac{n}{2}]} | y_{[\frac{n}{2}+1,n]}) = y \right] \\
&\qquad \text{(denote } g(x) \text{ by } y) \\
&= \Pr_{x \xleftarrow{\$} \{0,1\}^n} \left[f\left(h(y_{[\frac{n}{2}+1,n]}) \oplus y_{[1,\frac{n}{2}]} | y_{[\frac{n}{2}+1,n]} \right) \oplus \left(h(y_{[\frac{n}{2}+1,n]}) \oplus y_{[1,\frac{n}{2}]} | y_{[\frac{n}{2}+1,n]} \right) = y \right] \\
&= \Pr_{x \xleftarrow{\$} \{0,1\}^n} \left[\left(h(y_{[\frac{n}{2}+1,n]}) | 0^{\frac{n}{2}} \right) \oplus \left(h(y_{[\frac{n}{2}+1,n]}) \oplus y_{[1,\frac{n}{2}]} | y_{[\frac{n}{2}+1,n]} \right) = y \right] \\
&= \Pr_{x \xleftarrow{\$} \{0,1\}^n} \left[\left(h(y_{[\frac{n}{2}+1,n]}) \oplus h(y_{[\frac{n}{2}+1,n]}) \oplus y_{[1,\frac{n}{2}]} \right) \Big| \left(0^{\frac{n}{2}} \oplus y_{[\frac{n}{2}+1,n]} \right) = y \right] \\
&= \Pr_{x \xleftarrow{\$} \{0,1\}^n} \left[y_{[1,\frac{n}{2}]} | y_{[\frac{n}{2}+1,n]} = y \right] \\
&= \Pr_{x \xleftarrow{\$} \{0,1\}^n} [y = y] = 1.
\end{aligned}$$

This implies $\nu(n)$ is non-negligible, following from the conclusion of problem 1(i) and the fact that

$$\lim_{n \to \infty} \nu(n) \cdot n^c = \lim_{n \to \infty} 1 \cdot n^c = 1 \text{ or } \infty \neq 0$$

for any fixed integer $c \in \mathbb{N}$. This proves that $\mathbf{Adv}^\bigstar$ runs in PPT and inverts g with non-negligible probability, yielding that g is not one-way. ■

(v). We show g is not always one-way by providing a counterexample, i.e., a length-preserving one-way function f such that the corresponding g is not one-way.

Let h be an arbitrary length-preserving one-way function (which exists by the assumption that a length-preserving one-way function f exists) and c be an arbitrary real number in $\left(\frac{1}{4}, \frac{1}{2}\right)$ (say,

$c \triangleq \dfrac{1.14514}{4}$). We define our $f : \{0,1\}^* \to \{0,1\}^*$ as

$$f(x) \triangleq \begin{cases} 0^n & \text{if } \dfrac{|x|_1}{n} < c \\ 1|h(x_{[2,n]}) & \text{otherwise} \end{cases}$$

where we denote $n \triangleq |x|$ and define $|\cdot|_1$ as the number of 1's in a bit string, i.e.,

$$|b_1 b_2 \dots b_n|_1 \triangleq \sum_{i=1}^{n} b_n \quad \text{for each } x = b_1 b_2 \dots b_n \in \{0,1\}^*.$$

We claim that f is length-preserving and one-way, with a detailed proof deferred until later (see claim 4.3). Next, we claim that g is not one-way given the f we've just defined, which is supposed to be length-preserving and one-way. Likewise, we postpone the detailed proof of g being non-one-way until later (see claim 4.4).

Before verifying the (non-)one-wayness of f and g, we make some observations that come in handy later. Notice that given an n-bit string $x = B_1 B_2 \dots B_n \xleftarrow{\$} \{0,1\}^n$ sampled uniformly, each bit of x is in fact an independent, Bernoulli distributed random variable B_i with probability $p = \frac{1}{2}$, i.e., $B_1, B_2, \dots, B_n \overset{\text{iid}}{\sim} \operatorname{Ber}(\frac{1}{2})$. Therefore the Chernoff's bound says that

$$\begin{aligned} \mu_1(n) \triangleq \Pr_{x \xleftarrow{\$} \{0,1\}^n}\left[\frac{|x|_1}{n} < c\right] &= \Pr_{B_1,\dots,B_n \overset{\text{iid}}{\sim} \operatorname{Ber}(\frac{1}{2})}\left[\frac{1}{n}\sum_{i=1}^{n} B_i < c\right] \\ &\le \Pr_{B_1,\dots,B_n \overset{\text{iid}}{\sim} \operatorname{Ber}(\frac{1}{2})}\left[\left|\frac{1}{n}\sum_{i=1}^{n} B_i - \frac{1}{2}\right| \ge \frac{1}{2} - c\right] \\ &\le 2e^{-n(\frac{1}{2}-c)^2}. \end{aligned} \tag{🥺}$$

Since $\lim\limits_{n\to\infty} 2e^{-n(\frac{1}{2}-c)^2} n^{c'} = 0$ for any $c' \in \mathbb{N}$ and thus $2e^{-n(\frac{1}{2}-c)^2}$ is negligible (by problem 1(i)), by lemma 4.2 it follows that μ_1 is negligible.

Moreover, given 2 uniformly and independently sampled n-bit strings $x_1 = B_1 B_2 \dots B_n, x_2 = C_1 C_2 \dots C_n \xleftarrow{\$} \{0,1\}^n$, we know that $B_1, \dots, B_n, C_1, \dots, C_n \overset{\text{iid}}{\sim} \operatorname{Ber}(\frac{1}{2})$. Then each bit of the bitwise AND $x_1 \wedge x_2 = D_1, \dots, D_n$ of x_1 and x_2 is in fact an independent, Bernoulli distributed random variable D_i with probability $p = \frac{1}{4}$ since $\Pr[D_i = 1] = \Pr[B_i \wedge C_i = 1] = \Pr[B_i = 1]\Pr[C_i = 1] = \dfrac{1}{2} \cdot \dfrac{1}{2} = \dfrac{1}{4}$; equivalently, $D_1, D_2, \dots, D_n \overset{\text{iid}}{\sim} \operatorname{Ber}(\frac{1}{4})$. Likewise, the Chernoff's bound hence says that

$$\begin{aligned} \mu_2(n) \triangleq \Pr_{x_1, x_2 \xleftarrow{\$} \{0,1\}^n}\left[\frac{|x_1 \wedge x_2|_1}{n} < c\right] &= \Pr_{D_1,\dots,D_n \overset{\text{iid}}{\sim} \operatorname{Ber}(\frac{1}{4})}\left[\frac{1}{n}\sum_{i=1}^{n} D_i < c\right] \\ &\ge \Pr_{D_1,\dots,D_n \overset{\text{iid}}{\sim} \operatorname{Ber}(\frac{1}{4})}\left[\left|\frac{1}{n}\sum_{i=1}^{n} D_i - \frac{1}{4}\right| < c - \frac{1}{4}\right] \\ &= 1 - \Pr_{D_1,\dots,D_n \overset{\text{iid}}{\sim} \operatorname{Ber}(\frac{1}{4})}\left[\left|\frac{1}{n}\sum_{i=1}^{n} D_i - \frac{1}{4}\right| \ge c - \frac{1}{4}\right] \\ &\ge 1 - 2e^{-n(c-\frac{1}{4})^2}, \end{aligned} \tag{🤨}$$

which is non-negligible as 1 is clearly non-negligible and $2e^{-n(c-\frac{1}{4})^2}$ has been shown to be negligible. Thus, from lemma 4.1 it follows that μ_2 is non-negligible.

> Observation 4.2. μ_1 is negligible while μ_2 is non-negligible.

It remains to actually show that (1) f is length-preserving and one-way and that (2) g is not one-way given f.

Claim 4.3. f is a length-preserving one-way function.

Proof. Firstly, it's immediate that f is length-preserving as h is as well. Notice that f can be computed in PPT since h is one-way and thus can be computed in PPT by definition. Then, we prove f is one-way via reduction.

Suppose to the contrary that f is not one-way. As f can be computed in PPT, it must be that there exists a PPT algorithm $\mathbf{Adv}$ such that

$$\varepsilon(n) \triangleq \Pr_{x \xleftarrow{\$} \{0,1\}^n} [f(\mathbf{Adv}(1^n, f(x))) = f(x)]$$

is non-negligible. Then, we construct another PPT algorithm $\mathbf{Adv}^*$ that inverts h with non-negligible probability. Since h is one-way by assumption, this is a contradiction.

We construct $\mathbf{Adv}^*$ as follows: $\mathbf{Adv}^*$ takes as input 1^m and $y \coloneqq h(x)$, where $x \leftarrow \{0,1\}^m$. Then, it runs $\mathbf{Adv}$ on input $(1^{m+1}, 1|y)$, receiving as output x', and finally outputs $x'_{[2,m+1]}$. Namely,

$$\mathbf{Adv}^*(1^m, y) \triangleq \mathbf{Adv}(1^{m+1}, 1|y)_{[2,m+1]}.$$

It suffices to verify that $\mathbf{Adv}^*$ is a PPT algorithm that inverts h with non-negligible probability. Firstly, it's clear that $\mathbf{Adv}^*$ runs in PPT as it simply concatenates bit strings, runs $\mathbf{Adv}$ on an $(m+1)$-bit input, which also runs in PPT, and then splits a bit string. For the second part, we claim

$$\Pr_{x \xleftarrow{\$} \{0,1\}^m} [h(\mathbf{Adv}^*(1^m, h(x))) = h(x)]$$

is non-negligible.

■

cont'd. To see this, observe that for any $m \in \mathbb{N}$

$$\begin{aligned}
\Pr_{x \xleftarrow{\$} \{0,1\}^m}[h(\mathbf{Adv}^*(1^m, h(x))) = h(x)] &= \Pr_{x \xleftarrow{\$} \{0,1\}^m}\left[h(\mathbf{Adv}(1^{m+1}, 1|h(x))_{[2,m+1]}) = h(x)\right] \\
&= \Pr_{x^* \xleftarrow{\$} \{0,1\}^{m+1}}\left[h(\mathbf{Adv}(1^{m+1}, 1|h(x^*_{[2,m+1]}))_{[2,m+1]}) = h(x^*_{[2,m+1]})\right] \\
&\overset{\text{(a)}}{\geq} \Pr_{x^* \xleftarrow{\$} \{0,1\}^{m+1}}\left[f(\mathbf{Adv}(1^{m+1}, f(x^*))) = f(x^*) \wedge \frac{|x^*|_1}{m+1} \geq c\right] \\
&\overset{\text{(b)}}{\geq} \Pr_{x^* \xleftarrow{\$} \{0,1\}^{m+1}}\left[f(\mathbf{Adv}(1^{m+1}, f(x^*))) = f(x^*)\right] - \Pr_{x^* \xleftarrow{\$} \{0,1\}^{m+1}}\left[\frac{|x^*|_1}{m+1} < c\right] \\
&= \varepsilon(m+1) - \mu_1(m+1),
\end{aligned}$$

where inequality (a) comes from the fact that

$$\begin{aligned}
f(\mathbf{Adv}(1^{m+1}, f(x^*))) = f(x^*) \wedge \frac{|x^*|_1}{m+1} \geq c &\implies f(\mathbf{Adv}(1^{m+1}, 1|h(x^*_{[2,m+1]}))) = 1|h(x^*_{[2,m+1]}) \\
&\implies 1|h(\mathbf{Adv}(1^{m+1}, 1|h(x))_{[2,m+1]}) = 1|h(x^*_{[2,m+1]}) \\
&\iff h(\mathbf{Adv}(1^{m+1}, 1|h(x^*_{[2,m+1]}))_{[2,m+1]}) = h(x^*_{[2,m+1]}),
\end{aligned}$$

and inequality (b) we have used the union bounds proved in problem 3(i) (by taking set complement on both sides).

Since $\varepsilon(m+1)$ is non-negligible and $\mu_1(m+1)$ is negligible by observation 4.2, $\varepsilon(m+1) - \mu_1(m+1)$ is non-negligible. Therefore,

$$\Pr_{x \xleftarrow{\$} \{0,1\}^m}[h(\mathbf{Adv}^*(1^m, h(x))) = h(x)]$$

is also non-negligible. This proves that $\mathbf{Adv}^*$ runs in PPT and inverts h with non-negligible probability, contradicting the assumption that h is one-way. ■

Claim 4.4. g is not one-way given the f defined above.

Proof. To see this, we provide a PPT algorithm $\mathbf{Adv}^{\bigstar}$ that inverts g with non-negligible probability, i.e., that

$$\nu(n) \triangleq \Pr_{x \xleftarrow{\$} \{0,1\}^{2n}} \left[g(\mathbf{Adv}^{\bigstar}(1^{2n}, g(x))) = g(x) \right]$$

is non-negligible. We construct $\mathbf{Adv}^{\bigstar}$ as follows: $\mathbf{Adv}^{*}$ takes as input 1^{2n} and $y := g(x)$, where $x \leftarrow \{0,1\}^{2n}$. Then, it directly outputs (i.e. "guesses") 0^{2n}. Namely,

$$\mathbf{Adv}^{\bigstar}(1^n, y) \triangleq 0^{2n}.$$

It suffices to verify that $\mathbf{Adv}^{\bigstar}$ is a PPT algorithm that inverts g with non-negligible probability. Indeed, by construction $\mathbf{Adv}^{\bigstar}$ trivially runs in PPT. As for the second part, observe that for any $n \in \mathbb{N}$

$$\begin{aligned}
\nu(n) &\triangleq \Pr_{x \xleftarrow{\$} \{0,1\}^{2n}} \left[g(\mathbf{Adv}^{\bigstar}(1^{2n}, g(x))) = g(x) \right] \\
&= \Pr_{x \xleftarrow{\$} \{0,1\}^{2n}} \left[g(0^{2n}) = g(x) \right] \\
&= \Pr_{x \xleftarrow{\$} \{0,1\}^{2n}} \left[f(0^{n}) = g(x) \right] \\
&= \Pr_{x \xleftarrow{\$} \{0,1\}^{2n}} \left[0^{n} = g(x) \right] \\
&= \Pr_{x_1, x_2 \xleftarrow{\$} \{0,1\}^{n}} \left[0^{n} = f(x_1 \wedge x_2) \right] \\
&= \Pr_{x_1, x_2 \xleftarrow{\$} \{0,1\}^{n}} \left[\frac{|x_1 \wedge x_2|_1}{n} < c \right] \\
&= \mu_2(n),
\end{aligned}$$

which is non-negligible by observation 4.2. This proves that $\mathbf{Adv}^{\bigstar}$ runs in PPT and inverts g with non-negligible probability, yielding that g is not one-way. ∎

Now we've completed the proof. ∎

Reference

P1 (None)

P2
- Mathematical statistics (MATH4013) lecture notes (`topic_2.pdf`, `topic_3.pdf`)

P3
- Mathematical statistics (MATH4013) lecture notes (`topic_1.pdf`)

P4
- Examples of reductions for one-way functions by Noah Stephens-Davidowitz